

Basics of Spatial Regression

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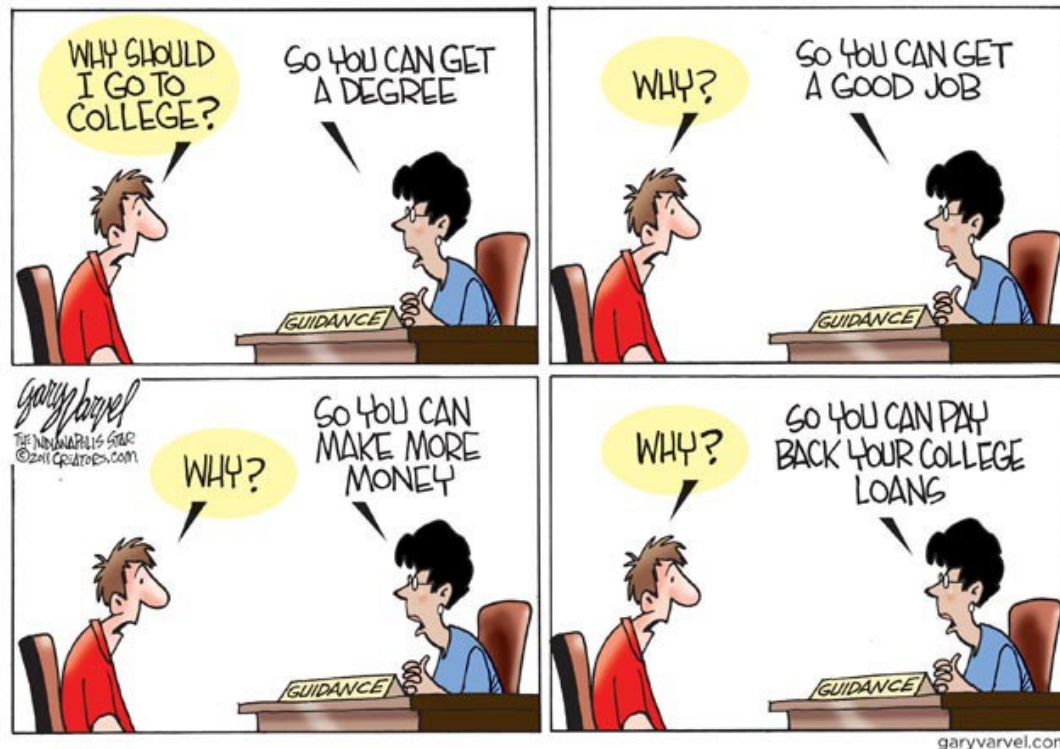
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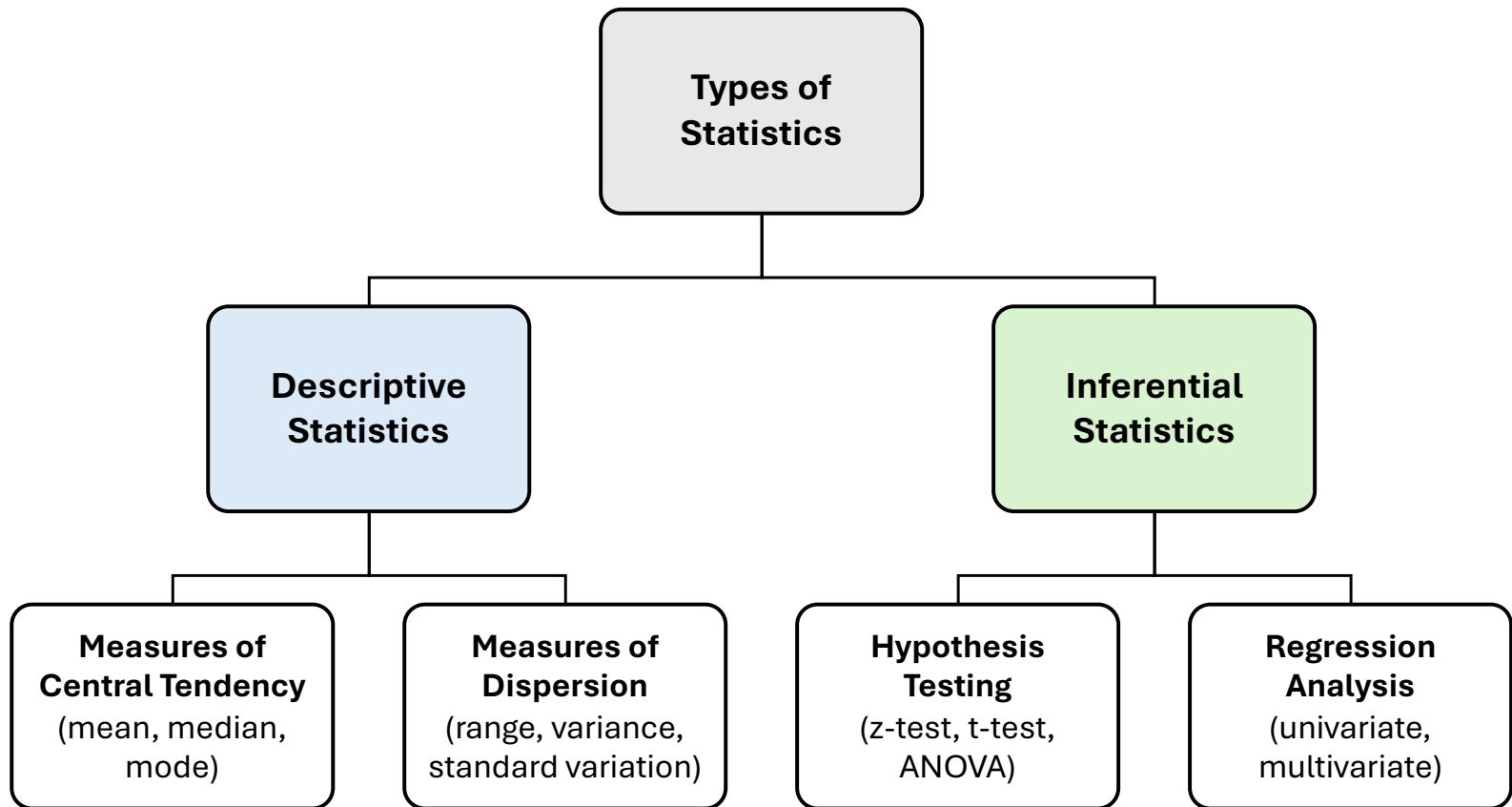


ENGINEERING
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Question

- We all know: *"Go to college, get a better job, make more money"*
- Let us look at all 254 Texas counties
- See if education is a driver of wealth

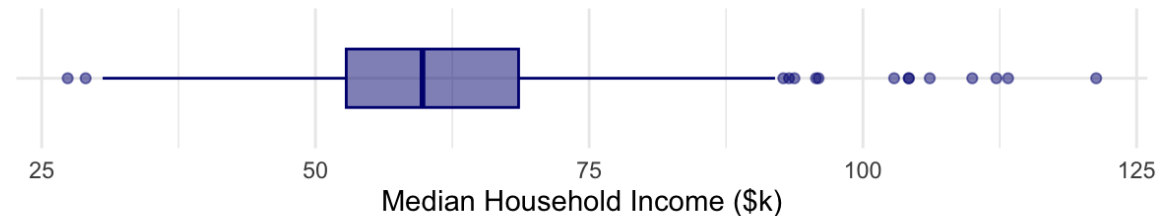




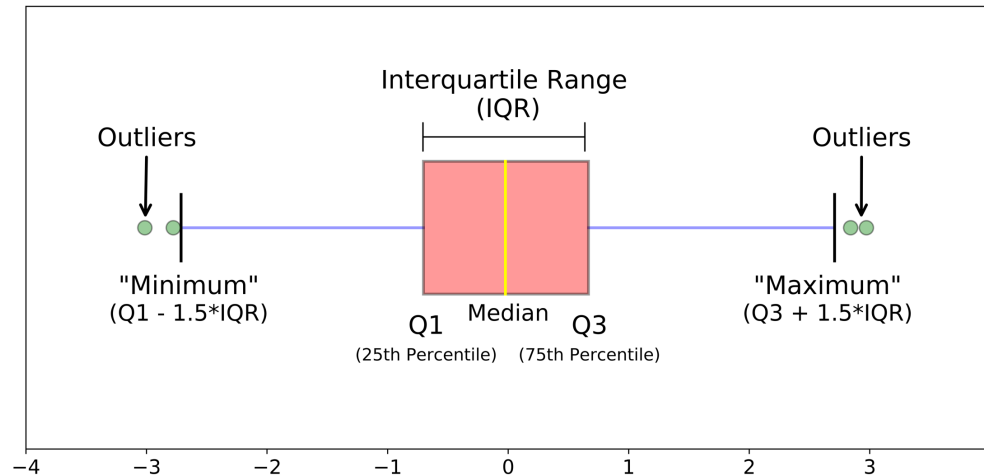
Data

- Pulled 2022 ACS 5-year census data.
- **Dependent Variable:** Median Household Income

Distribution of Median Household Income in the Past 12 Months
Texas Counties

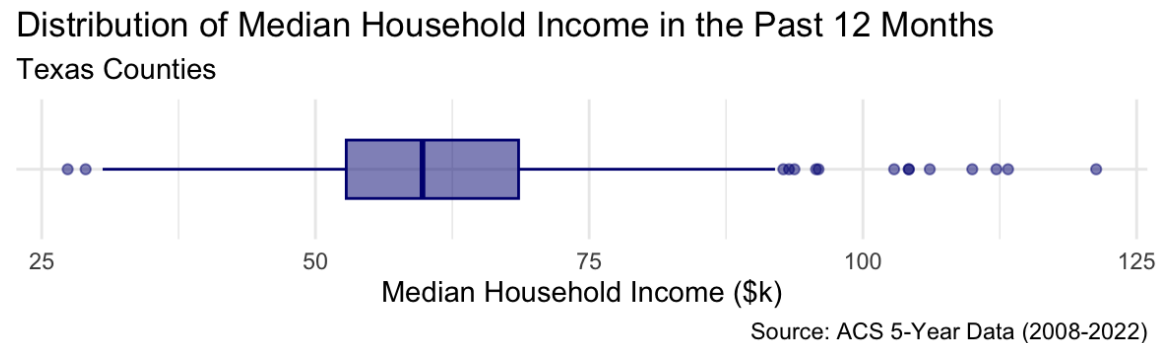


Source: ACS 5-Year Data (2008-2022)

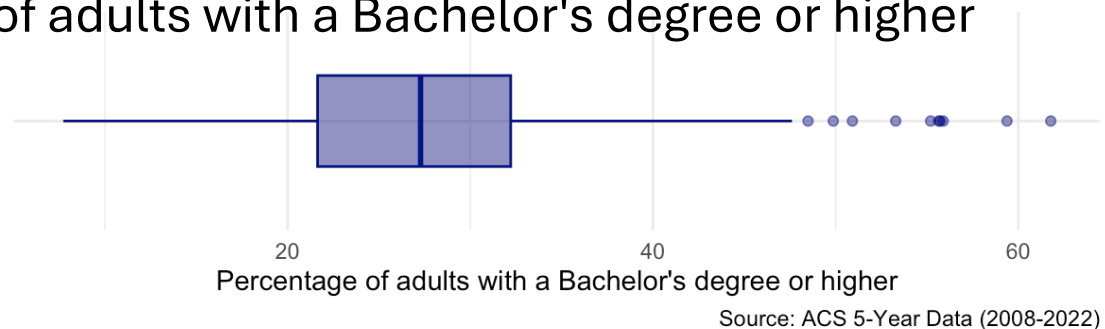


Data

- Pulled 2022 ACS 5-year census data.
- **Dependent Variable:** Median Household Income

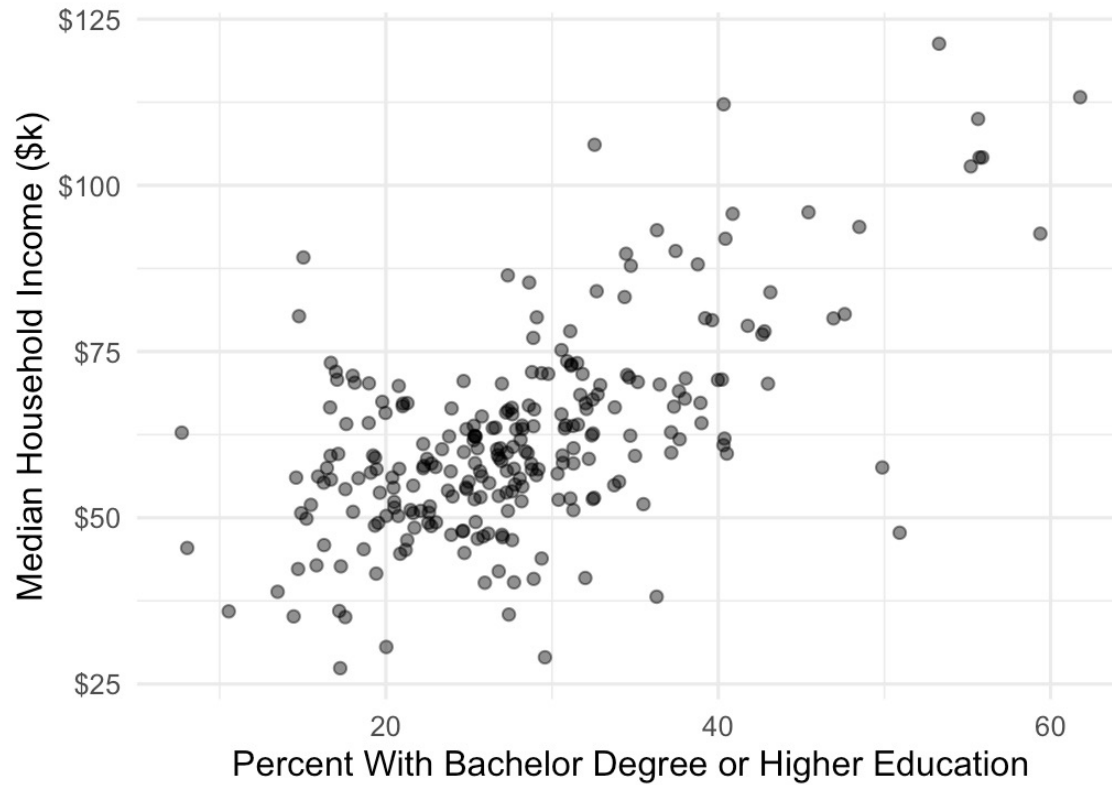


- **Independent Variable:**
Percentage of adults with a Bachelor's degree or higher



The Impact of Education on Household Income

Texas Counties



Source: ACS 5-Year Data (2008-2022)

Baseline (OLS)

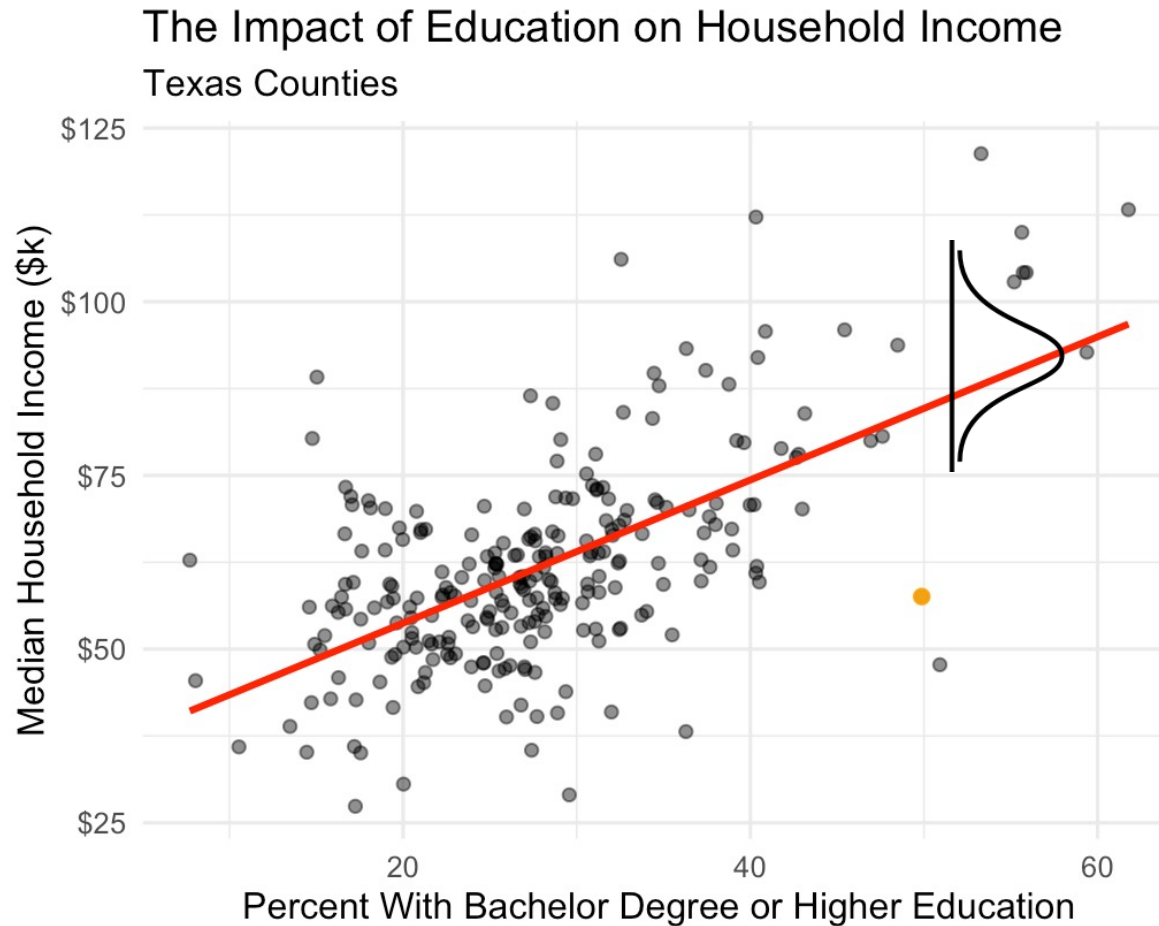
- First model, a simple linear regression:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

- Y: Median Household Income.
- X: % with a Bachelor's degree or higher.
- This model assumes ϵ (the error) is independent.

```
=====
                        Dependent variable:
-----
                        med_income_k
                        (1)                (2)
-----
pct_with_diploma                1.03***
Constant                62.04***        33.13***
-----
Observations                253                253
R2                0.00                0.38
Residual Std. Error 15.23 (df = 252)    11.98 (df = 251)
F Statistic                156.10*** (df = 1; 251)
=====
Note:                *p<0.1; **p<0.05; ***p<0.01
```

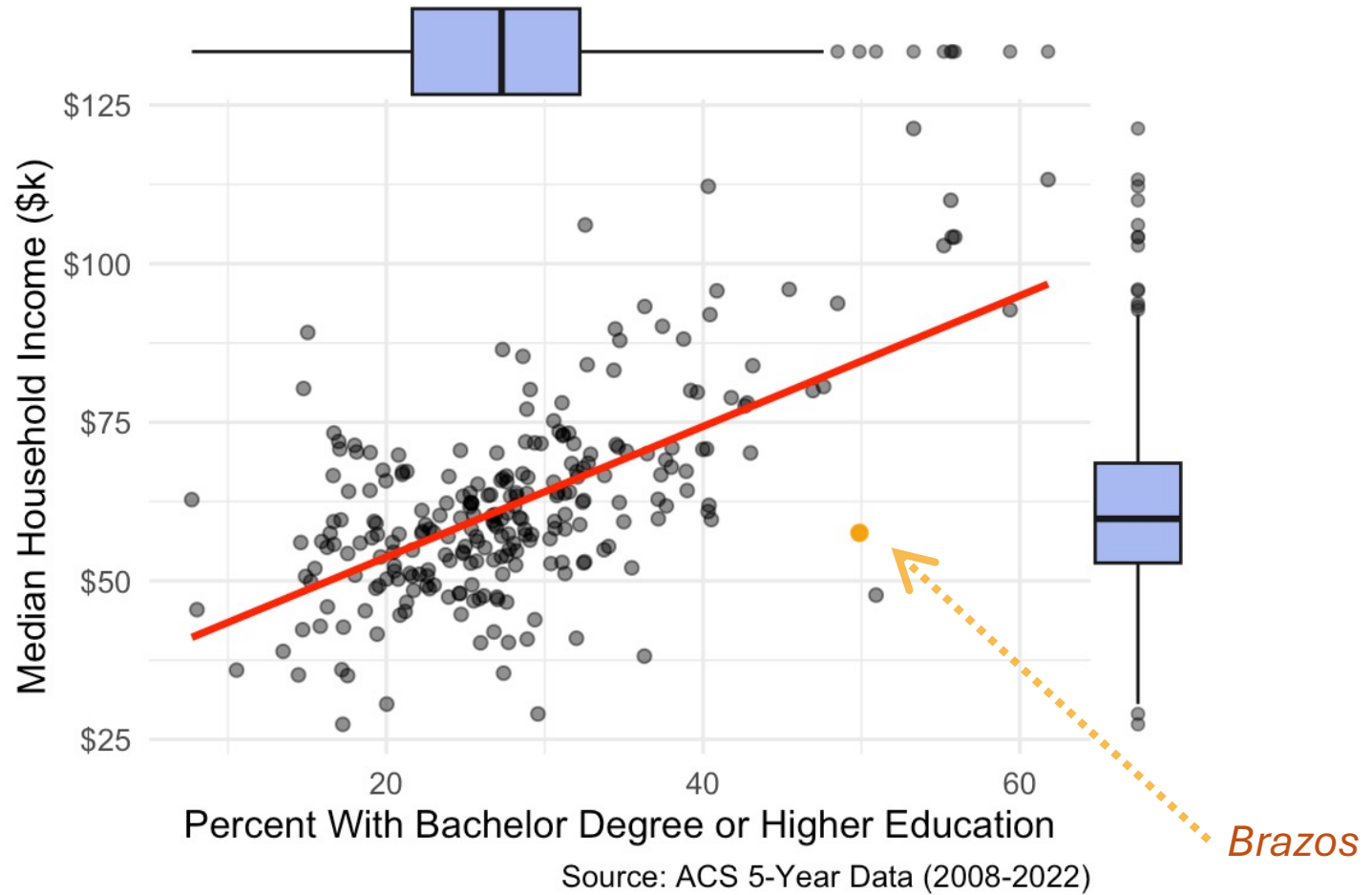
Baseline (OLS)



- As education goes up, income follows.
- Basic scatterplot and baseline model shows a strong, positive correlation
 - This model assumes every county is independent*
 - It assumes that what happens in Travis County has zero effect on Williamson County.
 - We know that's not how the real-world works.

- In a standard Ordinary Least Squares (OLS) regression, one of the core "Gauss-Markov" assumptions is that the observations (specifically the error terms) are independent of one another.
- When this assumption is met, the covariance between any two different error terms is zero: $Cov(\epsilon_i - \epsilon_j) = 0$ for $i \neq j$
- **Why Independence Matters?**
 - **Standard Errors:** If observations are correlated, OLS will typically underestimate the standard errors.
 - This leads to artificially low p-values and "false positives" (Type I errors)
 - **Reliability:** You might think a relationship is statistically significant when, in reality, you just have a bunch of repeating patterns in your data.
- Common Ways Independence is Violated
 - Autocorrelation (Time Series)
 - Clustering (Grouped Data)
 - Spatial Correlation

The Impact of Education on Household Income Texas Counties



- <https://www.bls.gov/ooh/fastest-growing.htm>
- <https://www.bls.gov/ooh/math/data-scientists.htm>
- <https://www.bls.gov/ooh/architecture-and-engineering/industrial-engineers.htm>

Data scientists

Outlook: +34% (Much faster than average)

Median pay: \$112,590

Jobs (2024): 246K

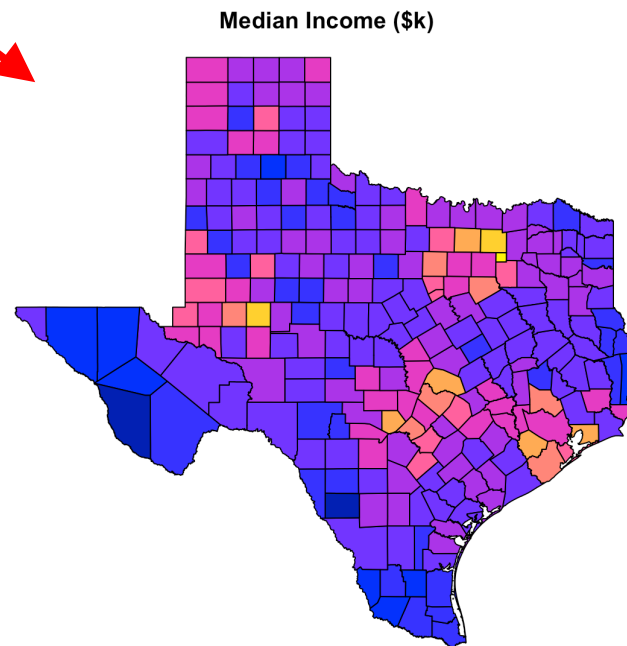
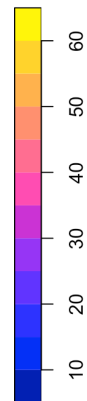
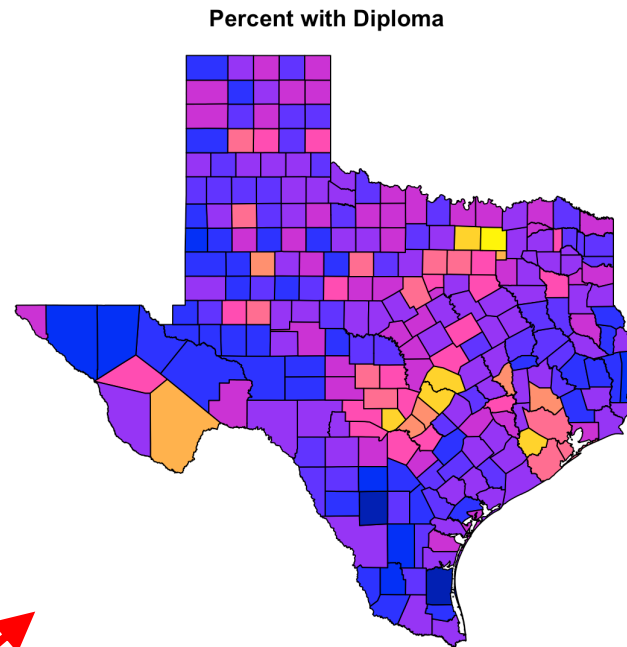
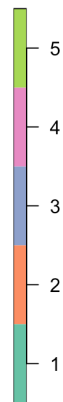
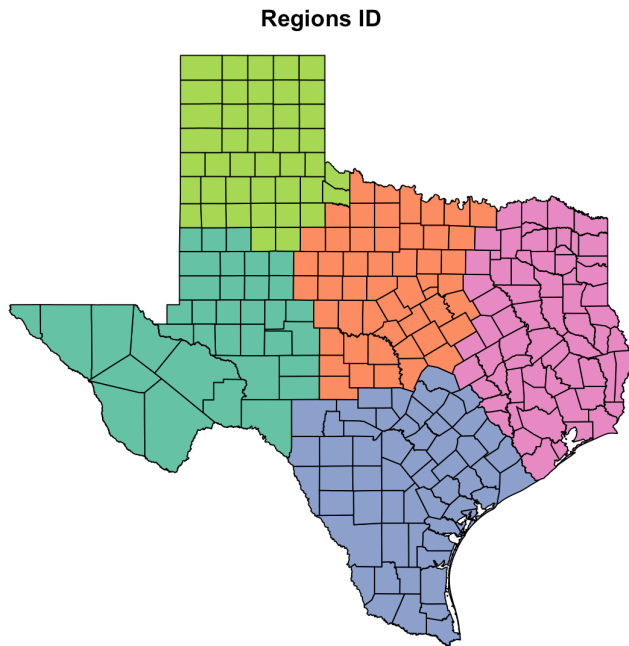
Outlook: 34% (Much faster than average)

Education: Bachelor's degree

Question 2

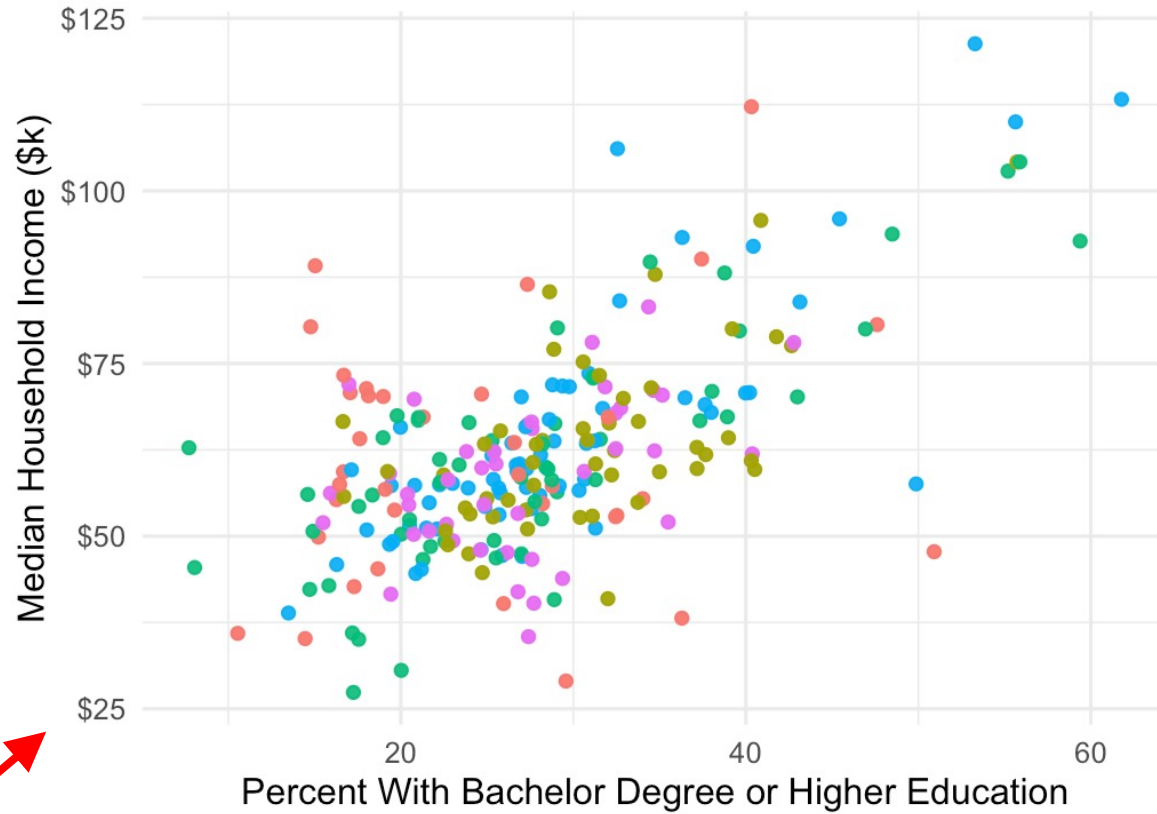
- Geography could also be an influencer too ...



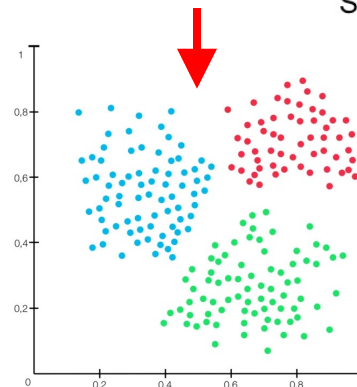
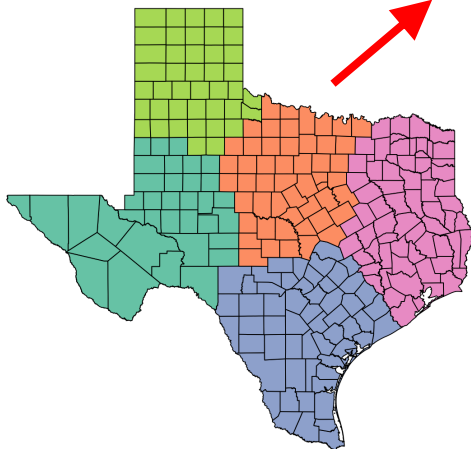


The Impact of Education on Household Income

Texas Counties



Source: ACS 5-Year Data (2008-2022)



Multilevel Model

- Texas is a patchwork of different economies (Valley, Permian Basin, High Plains, etc.)
- A degree in a rural region might yield a different income than a degree in a tech hub.
- We use a **Random Intercept Model** to account for regional nesting:

$$Y_i = \beta_0 + \beta_1 X_{ij} + u_j + \epsilon_{ij}$$

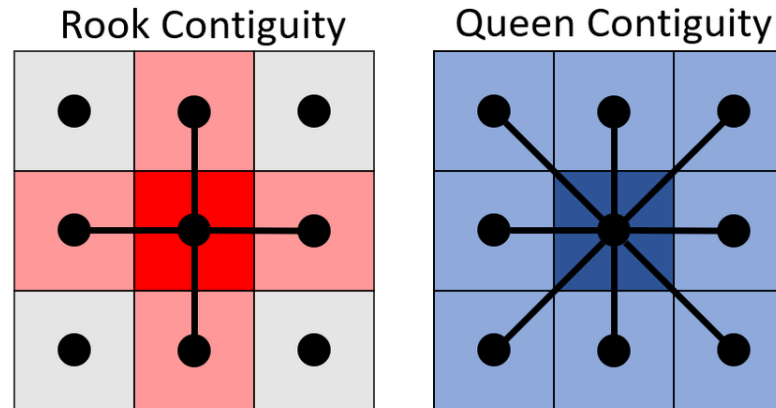
- We use K-means clustering to define 5 geographic regions (u_j)
- Goal: To see if the "Education Effect" stays the same once we acknowledge that different parts of Texas have different economic "baselines"

Multilevel Model

Dependent variable:			
	med_income_k		linear
	OLS		mixed-effects
	(1)	(2)	(3)
pct_with_diploma		1.03***	1.03***
Constant	62.04***	33.13***	33.10***
Observations	253	253	253
R2	0.00	0.38	
Adjusted R2	0.00	0.38	
Log Likelihood			-987.20
Akaike Inf. Crit.			1,982.00
Bayesian Inf. Crit.			1,997.00
Residual Std. Error	15.23 (df = 252)	11.98 (df = 251)	11.97
Note: *p<0.1; **p<0.05; ***p<0.01			

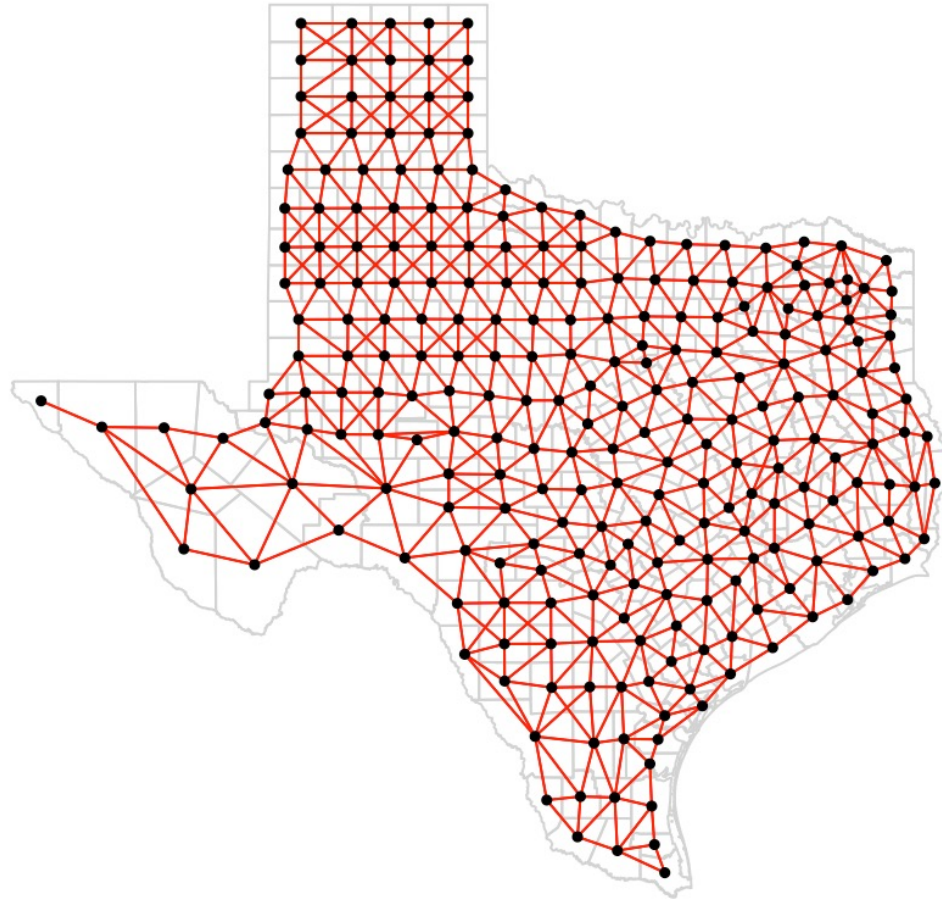
How do we define "nearby"?

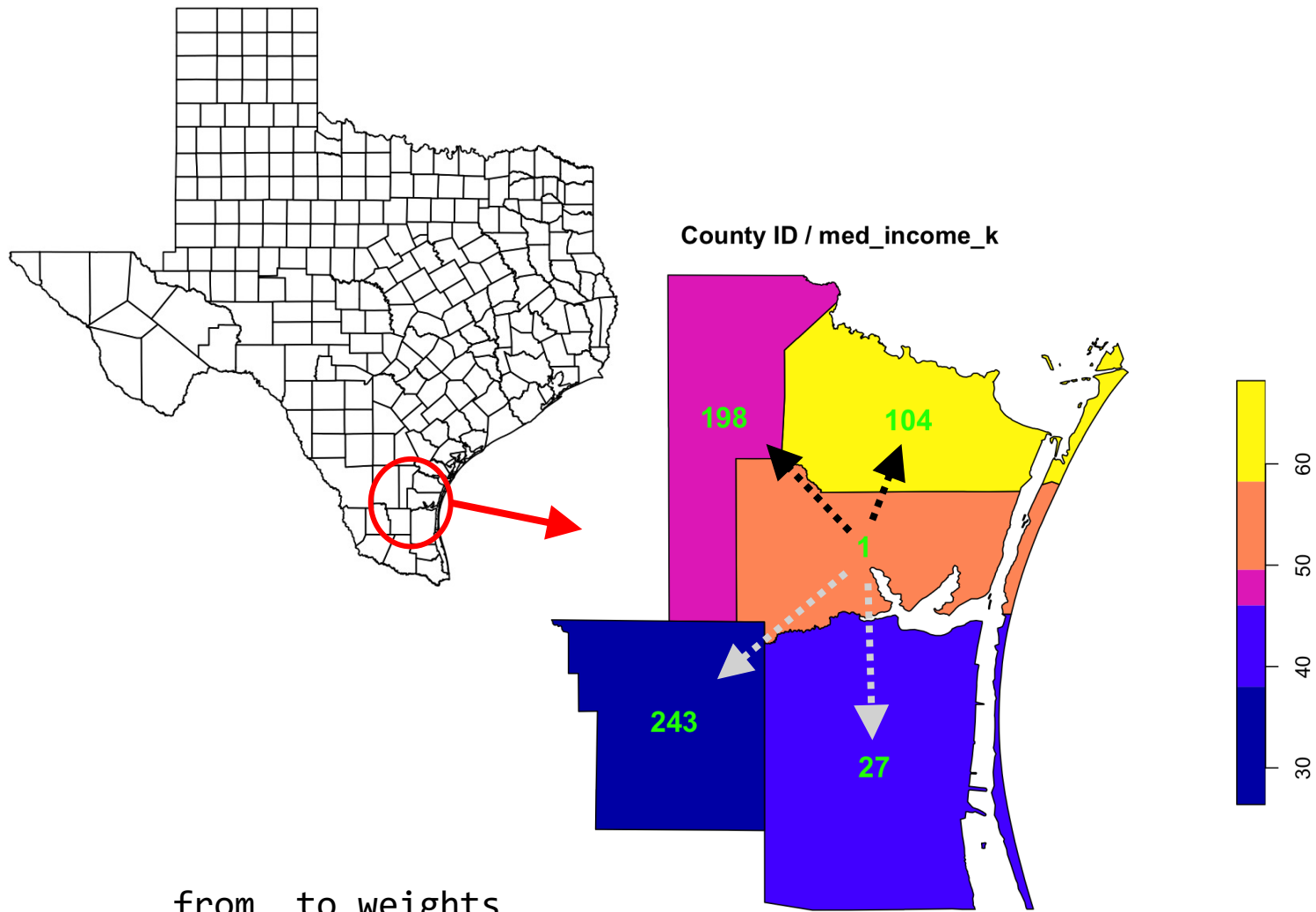
- In this analysis, let us use Queen Contiguity:
 - If two counties share even a single corner, they are neighbors (like a Queen on a chessboard)



Spatial Weights Matrix (W)

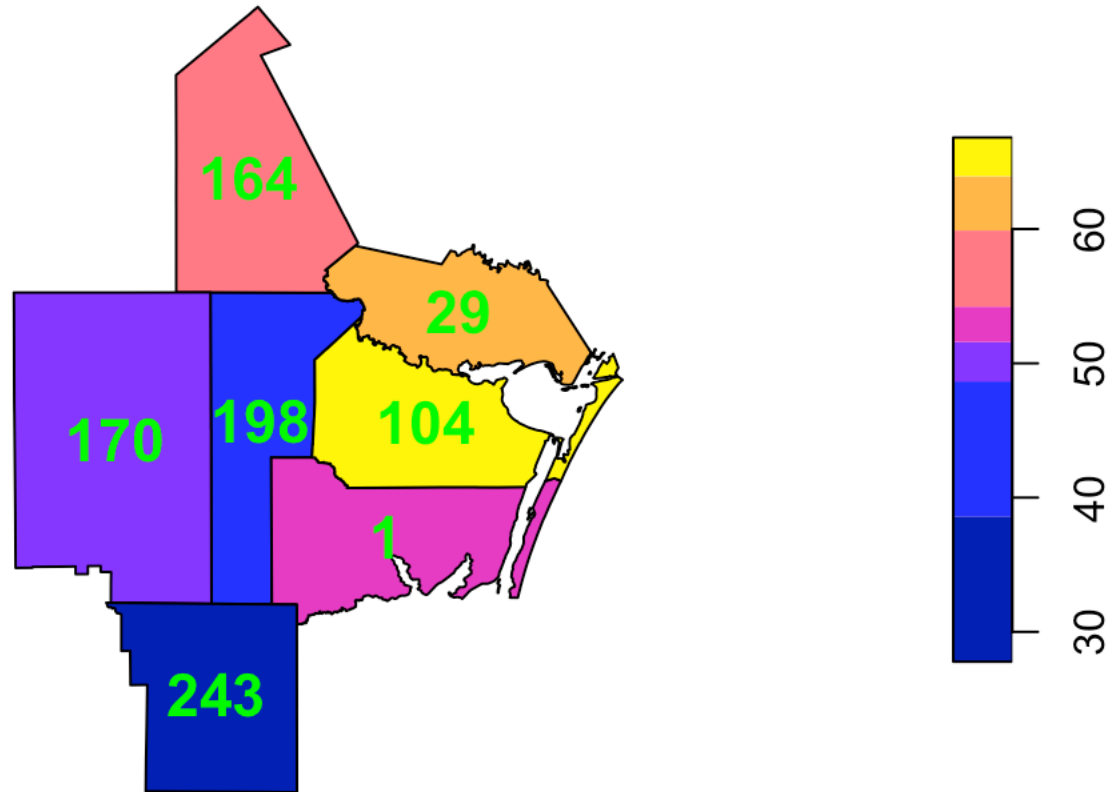
- If county A and B are neighbors, $W_{ab} = 1$; otherwise, 0
 - Then **row-standardize** the matrix so the influence of all neighbors sums to 1
 - This allows us to calculate a "Spatial Lag" essentially the weighted average of everything happening next by





	from	to	weights
1	1	27	0.25
2	1	104	0.25
3	1	198	0.25
4	1	243	0.25

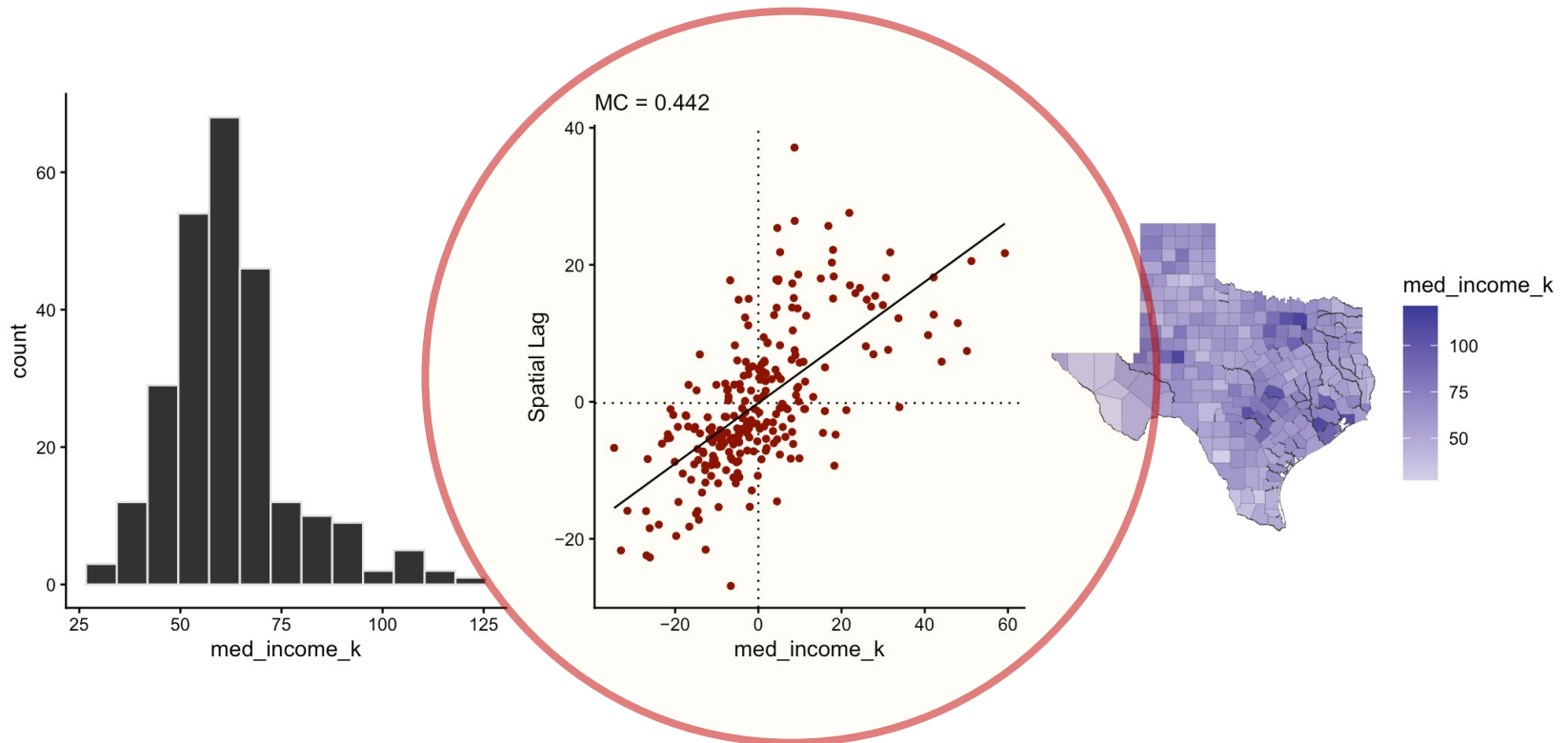
med_income_k



	from	to	weights
1	198	1	0.1667
2	198	29	0.1667
3	198	104	0.1667
4	198	164	0.1667
5	198	170	0.1667
6	198	243	0.1667

Spatial Lag

- The spatial lag (y-axis) represents the average value of the variable in neighboring locations for each observation.
 - The scatterplot axis are centered around 0.
 - **Mean-Centering:** The function calculates the average for all counties.
 - **Subtraction:** then subtracts that average from every county value ($x_i - \bar{x}$).
 - **Spatial Lag:** is the weighted average of the centered values of a county's neighbors.



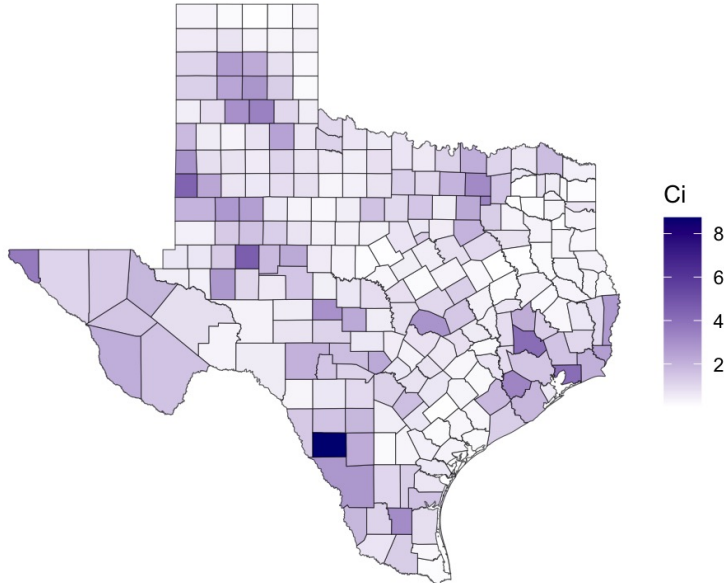
Local Indicators of Spatial Association (LISAs)

Local Geary: absolute difference between a county and its neighbors (dissimilarity).

Usually:

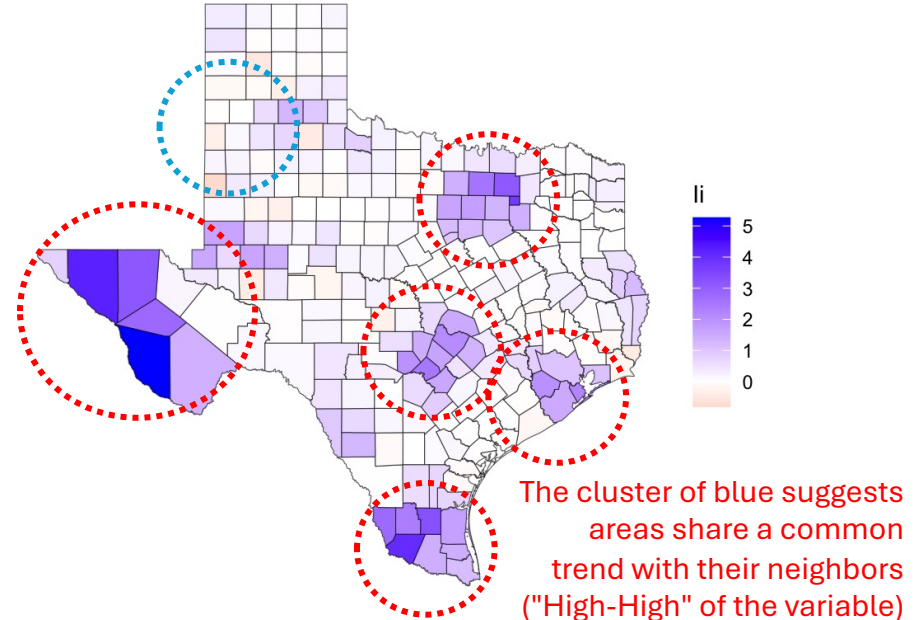
- **small** values of C_i indicate **positive spatial autocorrelation** (neighbors are similar)
- **large** values indicate **negative spatial autocorrelation** (neighbors are different)

The dark blue areas represent counties that have very different values compared to their immediate neighbors. These are your spatial outliers:



Local Moran's I: cross-product of deviations from the mean. Measures whether a county and its neighbors are both above or both below the average.

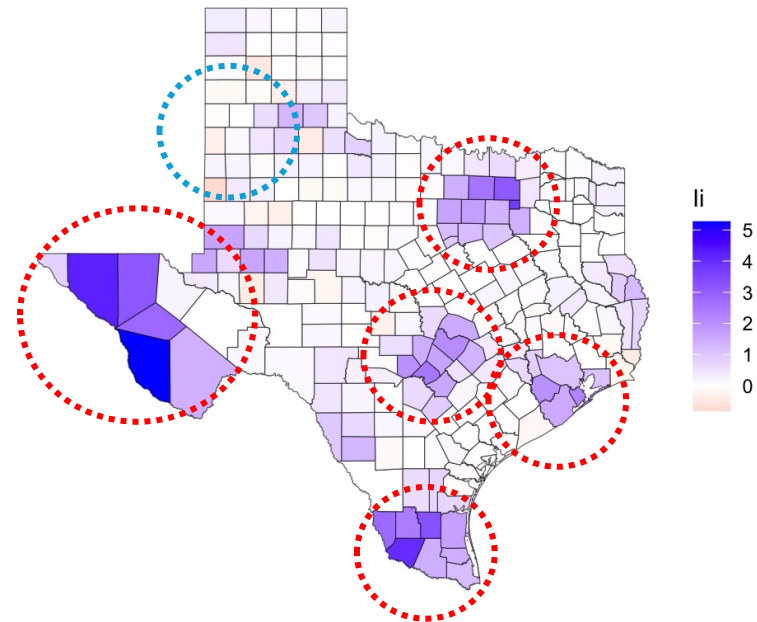
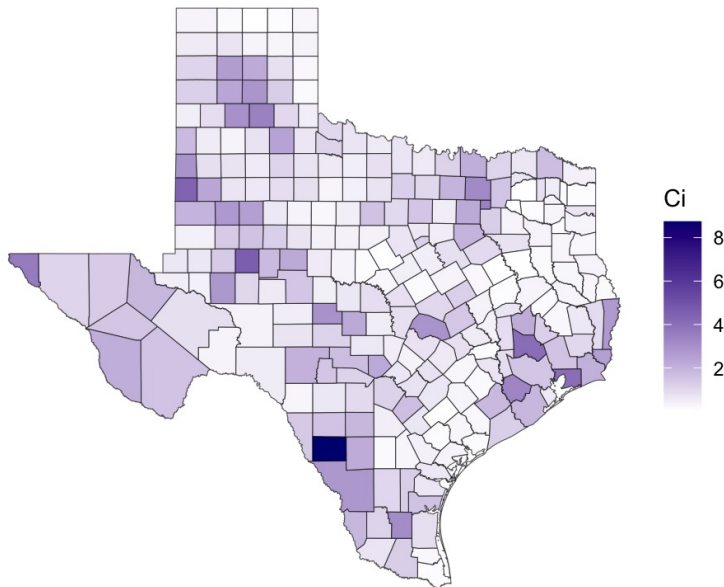
- **Positive values (Blue):** Indicate Spatial Clusters. This means the county is surrounded by neighbors with similar values (High-High or Low-Low).
- **Near Zero (White):** Indicates a random spatial distribution.
- **Negative values (Red/Pink):** Indicate Spatial Outliers. A "High" value surrounded by "Low" values, or vice versa.



Local Indicators of Spatial Association (LISAs)

- Indicates Clustering: specific spots surrounded by similar values

1. High-High (**Hot Spots**) A wealthy surrounded by wealthy
2. Low-Low (**Cold Spots**) A poor surrounded by poor
3. High-Low (Spatial Outlier)
4. Low-High (Spatial Outlier)



Spatial Lag Model (SAR)

- We are no longer just looking at $X^{(\text{Education})}$
we are looking at the *Spatial Lag* of Y :

$$Y = \rho WY + X\beta + \epsilon$$

- ρ (Rho): The autoregressive coefficient.
 - It tells us how much the income of neighboring counties "drags" your county's income up or down.
- WY : The "spatial lag": Represents the average income of the neighbors
- If ρ is significant, it means income is "contagious"

Multilevel Model

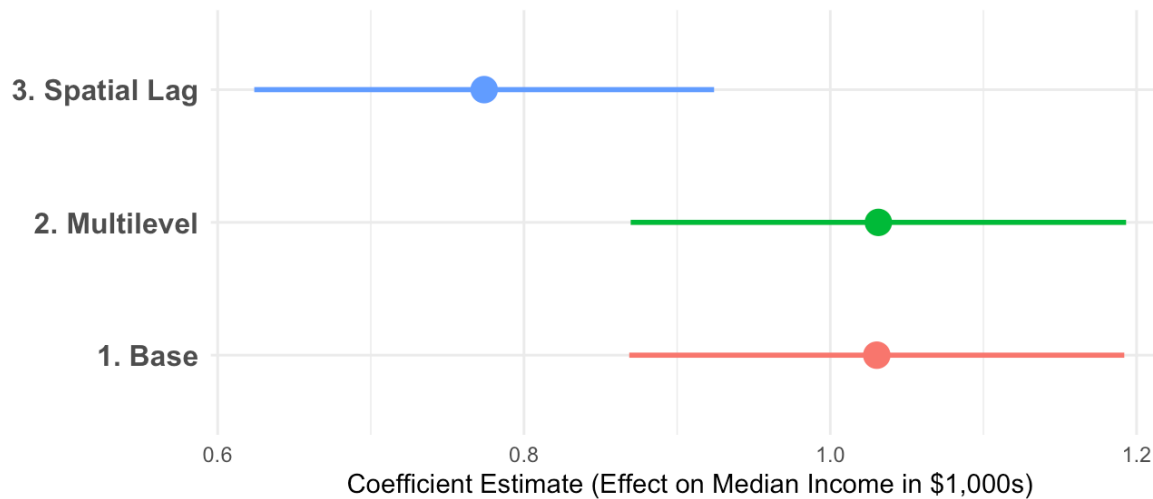
=====				
Dependent variable:				

	OLS		med_income_k	
			linear	spatial
	(1)	(2)	mixed-effects	autoregressive
			(3)	(4)

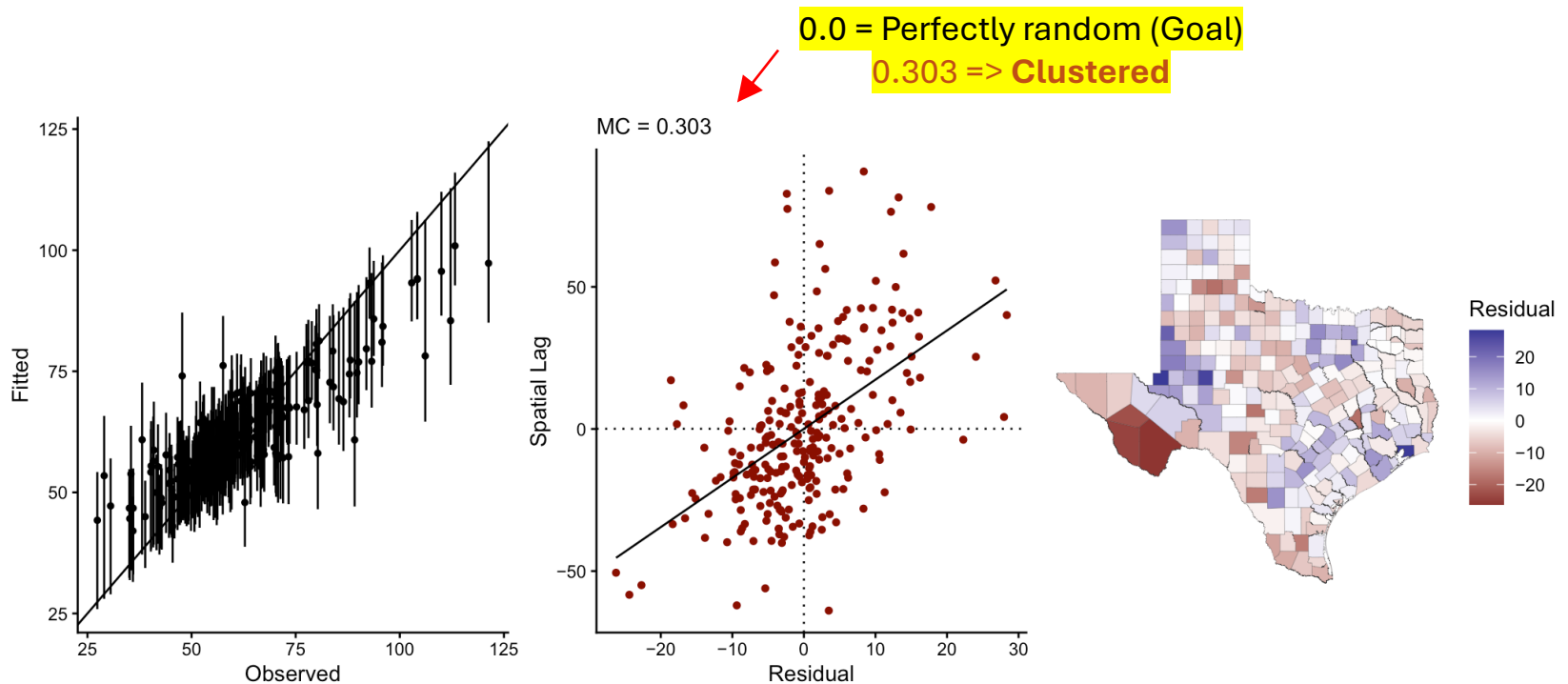
pct_with_diploma		1.03***	1.03***	0.77***
Constant	62.04***	33.13***	33.10***	7.58**

Observations	253	253	253	253
R2	0.00	0.38		
Adjusted R2	0.00	0.38		
Log Likelihood			-987.20	-953.20
Akaike Inf. Crit.			1,982.00	1,914.00
Residual Std. Error.	15.23 (df = 252)	11.98 (df = 251)	11.97	10.22
=====				
Note:			*p<0.1; **p<0.05; ***p<0.01	

Comparing the Effect of High Education on Income
Estimates and 95% Confidence Intervals



- In the simple OLS model:
 - "Education" was "stealing" the "credit" that actually belonged to the neighbors.
 - For instance: A county might be wealthy partly because of its degrees, but also because it's a bedroom community for a neighboring wealthy city.
- The Spatial Lag model separates these two:
 - The coefficient for Education gets smaller.
 - Tells us the direct effect of the degree versus the indirect effect of the location.
- Conclusions:
 - Education remains a primary driver of income across all models.
 - Being educated in a "wealthy cluster" is different than in a "spatial void".
 - Implication: To boost Texas economy, we can't just fund schools in isolation but foster regional ecosystems bc prosperity clearly spills over boundary lines.



One more thing ("Model Stress Test")

- Where is the spatial model succeeding?
Where is the "Geography of Texas" still outsmarting your math?
- Notice that as income gets higher (the right side of the graph), the error bars get longer and the dots drift further from the line.
- The model is good at predicting income for "average" counties, struggles with the extremely wealthy or extremely poor outliers.
 - 0.303 tells that the model errors are "contagious."
 - If the model under-predicts income in one county, it is very likely to under-predict it in the neighboring county too.
 - We have reduced the spatial bias, but we haven't eliminated it!

Learn More (Vignettes)

- Exploratory spatial data analysis
<https://cran.r-project.org/web/packages/geostan/vignettes/measuring-sa.html>
- Spatial analysis (with geostan)
<https://cran.r-project.org/web/packages/geostan/vignettes/spatial-analysis.html>
- Spatial measurement error models
<https://cran.r-project.org/web/packages/geostan/vignettes/spatial-me-models.html>
- Custom spatial models (with RStan and geostan)
<https://cran.r-project.org/web/packages/geostan/vignettes/custom-spatial-models.html>
- Raster regression
<https://cran.r-project.org/web/packages/geostan/vignettes/raster-regression.html>

References

- Spatial Data Science
<https://r-spatial.org/book/>
 - Spatial Regression (Markov random field and multilevel models)
<https://r-spatial.org/book/16-SpatialRegression.html>
 - Spatial Econometrics Models
<https://r-spatial.org/book/17-Econometrics.html>